

Chanhee Park 박찬희

AI Engineer — LLM Evaluation, Data, and Agent Systems

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What you can measure is what you end up training on. Benchmarks saturate while the tasks that matter — multi-turn, RAG, agents — go unmeasured, and LLM-as-a-judge replaced human judgment before anyone validated it. I build evaluation that holds up, then use it to decide what data survives and what gets trained.

EXPERIENCE

Tech Lead, AI Engineer (freelance) · LOUM Partner

Oct 2025 – Present · Seoul

loupnlp.com — LLM platform for HR and labor-law workflows, led by a practicing labor attorney (HLI, Seongnam). Four-person team.

MVP Jan 2026; live beta.

- **Built the eval harness** — 5-dimension LLM-as-judge over 1,000+ practitioner-sourced queries; ablation A/B across composer/retrieval/preset arms; model-migration gates. Custom depth-delivery metric caught **deep-query** → **generic-answer** failures invisible to aggregate scoring.
- **Planner / Executor / Composer architecture**, replacing a single-orchestrator design that collapsed under prompt overload (tool inventory, intent classification, IRAC synthesis over 50+ documents, multi-turn state in one context). Small fast router, parallel tool executor, heavy synthesis model. Response time **2.5 min** → **40 s** fast path, **~1 m 20 s** document-heavy.
- **Legal retrieval stack** — 9 public Korean legal databases (statutes, case law, administrative rules), live web search, attorney-curated documents; legal-fine-tuned KURE embeddings.
- **Shipped** document generation, review, consulting; multi-turn sessions on S3; EC2, Lambda, CloudWatch. **1.1K users** in closed beta.

Research Intern → Graduate Researcher · NLP&AI Lab, Korea University

Oct 2024 – Present · Seoul

- **Funnel of Thoughts** — inference-time algorithm preserving 32-rollout self-consistency accuracy at half the attention FLOPs: **56.1% fewer generation FLOPs, 37.6% less wall time**, no retraining, reward model, or logit access. Transfers to held-out architectures and out-of-domain tasks without retuning.
- Signal found by analysing **115K reasoning trajectories** (~0.8B tokens, 6 models, AIME24/25, AMC23, MATH500): hesitation-marker density predicts unproductive rollouts at zero cost.
- **AgentHop** — 1,011-item diagnostic benchmark for agentic multi-hop scientific QA over an arXiv citation graph (7,205 papers, 450K+ edges); 7-tool sandbox, fixed token/turn/tool-call budgets. Four-axis decomposition (retrieval, synthesis, tool-use, resource management) localizing agent failure; **19 models evaluated**. 8-stage construction pipeline with 3-model ensemble filtering and 7-auditor human review.

Interpreting Officer (First Lieutenant) · Republic of Korea Air Force — Logistics Command

Mar 2020 – Jun 2023 · Daegu

- Lead interpreter, defense procurement with Boeing, BAE Systems, Lockheed Martin; quadrilateral ROKAF-USAF-DAPA-Boeing talks, St. Louis. ROK-US Combined Exercises; Black Eagles airshow tour, 4 countries.

FIRST-AUTHORED PAPERS

AgentHop: A Diagnostic Benchmark for Agentic Multi-Hop Scientific Question Answering. C. Park, J. Yoon, S. Han, H. Moon, H. Lim. *NeurIPS 2026 Evaluations & Datasets* — under review

Funnel of Thoughts: Efficient Test-Time Scaling via Early Voting and Rollout Pruning. C. Park, S. Han, J. Yoon, S. Hong, H. Lim. *ACL Rolling Review 2026* — under review

MIRAGE: A Metric-Intensive Benchmark for Retrieval-Augmented Generation Evaluation. C. Park, H. Moon, H. Lim. *Findings of NAACL 2025* — 19 citations

Can Small Open-Source Language Models With Retrieval-Augmented Generation Match GPT-4 Performance in Breast Cancer Clinical Decision Support? C. Park, I. H. Park, M. Kim, Y. Jang, H. Lim, Y. Hur. *JCO Clinical Cancer Informatics*, 2026

PATENTS & SOFTWARE REGISTRATION

Evaluating retrieval-augmented generation systems (MIRAGE) — filed KR, Dec 2025. **Accelerating multi-sample inference of large reasoning models (FoT)** — filed KR, Aug 2026.

FUNDED PROJECTS

Sovereign AI Foundation Model · NC AI — SFT cold-start data for **VAETKI-100B-A10B** — 60+ open datasets plus AIHub and in-house sources 4 languages and 9 domains, curated to **6.5M Q/A samples** via 6-dimension clarity scoring, CoT-necessity filtering, and consistency-triggered human review (Sep–Dec 2025)

Korean Writing & Grammar Assistant · KOCCA — 3-pillar agent architecture; evaluation metric design, proposal authoring (Aug 2026 – present)

LegalMi:dm · KT — continual pretraining on Mi:dm-2.0-Base (11.5B); data curation, DeepSpeed multi-node (Jul 2024 – Jul 2025)

Clinical RAG for Breast Cancer Decision Support · Korea University internal — led end-to-end; published in *JCO Clin. Cancer Informatics* (May–Dec 2024)

TECHNICAL SKILLS

Serving vLLM, SGLang, Amazon SageMaker

Training PyTorch, Transformers, DeepSpeed, TRL — full fine-tuning, PEFT, and preference optimization (DPO variants) on multi-GPU single-node; multi-node with Ray

Infra Python, AWS (EC2 serving, S3, Lambda, CloudWatch), Docker, MongoDB

Languages Korean (native) · English (professional; TOEIC 980) · German (professional; TestDaF 4/5/4/5)

EDUCATION

M.S., Computer Science and Engineering · Korea University

Feb 2025 – Feb 2027 (expected) · Seoul

NLP&AI Lab, advised by Prof. Heuiseok Lim. Agent evaluation, efficient inference, domain adaptation.

B.A., Korean Language and Literature · Korea University

Mar 2014 – Feb 2020 · Seoul

Second major, German Literature. Exchange: Univ. of Bonn, Germany (2017–18).